EPIGO Math Notes

Entropy and Spectral Structure of Epigenomic Tracks

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This document describes the mathematical foundations underlying the EPIGO analysis pipeline. The focus is on the formal representation of epigenomic signal tracks and the two core computations applied to them: **Shannon entropy** (`shannon_entropy`) and a **one-dimensional spectrum** (`spectrum_1d`). The presentation emphasizes standard, implementation-agnostic definitions that correspond directly to the numerical operations performed by the pipeline.

1 Signal model (what a "track" is mathematically)

Epigenomic signal tracks (e.g. Roadmap MACS2 p-value signal[1])are modeled as *genome-indexed scalar fields*:

$$x : \{(c, p)\} \to \mathbb{R}_{\geq 0}, \tag{1}$$

where c denotes a chromosome identifier and p denotes a genomic coordinate (measured in base pairs). The value $x(c, p)$ represents the signal intensity at position p on chromosome c.
In practice, these signals are stored in interval-based formats. In bedGraph, the signal is represented as a *piecewise-constant* function:

$$x(p) = v_i \quad \text{for } p \in [s_i, e_i), \tag{2}$$

where each row (c, s_i, e_i, v_i) specifies that the signal takes the constant value v_i on the half-open interval $[s_i, e_i)$ of chromosome c.
[2]

For numerical analysis, the interval representation is converted into a uniformly sampled discrete signal by binning genomic positions into fixed-width bins of size B base pairs. For bin index k, the binned signal value is defined as

$$x_k \;=\; \frac{1}{B} \int_{kB}^{(k+1)B} x(p)\, dp \quad \approx \quad \frac{1}{B} \sum_{p=kB}^{(k+1)B-1} x(p). \tag{3}$$

This operation computes the average signal intensity over each bin, producing a vector $x = (x_0, \ldots, x_{N-1})$ that serves as the discrete representation of the track over a region, chromosome, or concatenated genome.

[1] Roadmap Epigenomics provides a uniformly processed reference collection of epigenomic maps; the flagship integrative analysis describes the 111 reference epigenomes resource [7], with an accompanying Nature editorial overview [9]. For this project, consolidated MACS2 p-value signal tracks were sourced from the Roadmap directory of p-value bigWig tracks [2]. MACS-derived p-values reflect statistical significance of enrichment over background in the MACS framework [11].

[2] The Roadmap tracks are distributed as bigWig, a compressed indexed binary format designed for efficient browsing and remote access to dense genomic signal data [1,5]. For numerical processing, bigWig is commonly converted to bedGraph using UCSC utilities; in this project the conversion toolchain is referenced via the Bioconda documentation entry for `bigWigToBedGraph` [3].

Normalization. Entropy is defined on probability distributions. To obtain a discrete distribution from the binned signal, intensities are normalized as

$$p_k \;=\; \frac{x_k + \epsilon}{\sum_{j=0}^{N-1}(x_j + \epsilon)}, \qquad \epsilon > 0. \tag{4}$$

Here p_k represents the fraction of total signal mass assigned to bin k. The additive constant ϵ ensures numerical stability by preventing zero-valued probabilities, which would otherwise lead to undefined logarithms.

2 Module: `analysis.py` (core math)

2.1 Shannon entropy (`shannon_entropy`)

Given a probability vector $p = (p_0, \ldots, p_{N-1})$ satisfying $p_k \geq 0$ and $\sum_k p_k = 1$, Shannon entropy is defined as

$$H(p) \;=\; -\sum_{k=0}^{N-1} p_k \log p_k. \tag{5}$$

[3]

In this context, entropy quantifies how evenly the signal mass is distributed across genomic bins. A high value indicates a diffuse signal spread over many bins, while a low value indicates concentration in a small subset of bins.
To enable comparison across regions with different numbers of bins, a normalized entropy is often used:

$$H_{\text{norm}}(p) \;=\; \frac{H(p)}{\log N}, \tag{6}$$

where N is the number of bins. This normalization bounds the entropy to the interval $[0, 1]$.

Robust computation. In practice, bins with exactly zero probability are omitted from the sum:

$$H(p) = -\sum_{k:p_k>0} p_k \log p_k. \tag{7}$$

This is equivalent to the full definition but avoids numerical issues. When entropy is computed from raw intensities, the probabilities p_k are obtained via the normalization described above.

2.2 1D spectrum (`spectrum_1d`)

To characterize spatial structure, the discrete signal x_k is analyzed in the frequency domain.

Detrending / centering. First, the mean signal level is removed:

$$\tilde{x}_k = x_k - \bar{x}, \quad \bar{x} = \frac{1}{N}\sum_{k=0}^{N-1} x_k. \tag{8}$$

This step removes the zero-frequency (DC) component, ensuring that subsequent spectral components reflect spatial variation rather than absolute signal level.

[3]Shannon introduced entropy as the fundamental measure of information and uncertainty for discrete probability distributions [8]. Modern textbook treatments and extensions are given in [4,6].

Discrete Fourier transform (DFT). The centered signal is transformed using the discrete Fourier transform:

$$X_m \;=\; \sum_{k=0}^{N-1} \tilde{x}_k \, e^{-i2\pi mk/N}, \qquad m = 0, \ldots, N-1. \tag{9}$$

Here m indexes spatial frequencies, and X_m measures the contribution of oscillations with period N/m bins.
The power spectrum is defined as

$$S_m \;=\; \frac{1}{N}\,|X_m|^2, \tag{10}$$

which represents the variance of the signal attributable to frequency m. Because $\tilde{x}_k$ is real-valued, the spectrum is symmetric and only the one-sided spectrum is typically retained.

Frequency and scale. If each bin spans B base pairs, the spatial frequency corresponding to index m is

$$f_m = \frac{m}{NB} \quad \text{(cycles per base)}, \tag{11}$$

with an associated characteristic wavelength $\lambda_m \approx 1/f_m$.

Optional smoothing. Raw periodograms exhibit high variance. To obtain smoother spectral estimates, adjacent frequency bins may be averaged:

$$\hat{S}_m = \sum_{r=-R}^{R} w_r \, S_{m+r}, \quad \sum_r w_r = 1. \tag{12}$$

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3 Module: `tracks.py` (track abstractions and derived features)

This module operates on discrete track representations and derived quantities:

1. Conversion of interval-based signals into binned vectors x.
2. Construction of feature vectors

   $$\phi(x) = \Big(H(p(x)),\ \hat{S}(x),\ \text{summary statistics}\Big), \tag{13}$$

   which combine information-theoretic and spectral descriptors.
3. Quantitative comparison of tracks via distances such as

   $$d(\phi_1, \phi_2) = \|\phi_1 - \phi_2\|_2. \tag{14}$$

Information geometry note. When tracks are interpreted as probability distributions, divergences such as the Kullback–Leibler divergence

$$D_{\mathrm{KL}}(p\|q) = \sum_k p_k \log \frac{p_k}{q_k} \tag{15}$$

induce a local information-geometric structure[5].

[4] Variance reduction via averaging modified periodograms is classically associated with Welch's method [10].
[5] Standard treatments of relative entropy and related information measures are given in [4, 6].

4 Module: `viz_tracks.py` and `viz.py` (visualization math)

Visualization routines display signal traces, entropy summaries, spectral estimates, and derived comparisons computed by the analysis and track modules.

5 Practical notes (stability and reproducibility)

Zero handling. All entropy calculations employ ϵ-smoothing to avoid undefined logarithms:

$$p_k \propto x_k + \epsilon.$$

Choice of bin size. The bin width B controls both entropy and spectral resolution: larger bins emphasize coarse-scale structure, while smaller bins increase resolution at the cost of higher variance.

Finite regions. When analysis is restricted to genomic windows, all definitions apply with N equal to the number of bins per window, and results may be aggregated across windows.

Summary. EPIGO represents epigenomic tracks as discrete signals, converts them into probability distributions for entropy analysis, and characterizes spatial structure using frequency-domain methods.

References

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